

606019



Test Booklet No.

ENGLISH

Test Booklet Code

03

NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

SYLLABUS:

Physics:	Electric charges and field, Electrostatic Potential and Capacitance, Current Electricity, Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction, Alternating Current, Electromagnetic Waves
Chemistry:	Solutions, Electrochemistry, Chemical Kinetics, p-Block Elements, d & f-Block Elements, Coordination Compounds
Botany:	Sexual reproduction in flowering plants, Principles of inheritance and variations, Molecular basis of inheritance
Zoology:	Human Reproduction, Reproductive Health, Evolution

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with blue/black ball point pen only.
- The test is of 3 hours 20 minutes duration and the Test Booklet contains 200 multiple-choice questions (four options with a single correct answer) from Physics, Chemistry and Biology (Botany and Zoology). 50 questions in each subject are divided into two Sections (A and B) as per details given below:
 - Section A shall consist of 35 (Thirty-five) Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
 - Section B shall consist of 15 (Fifteen) questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to attempt any 10 (Ten) questions out of 15 (Fifteen) in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is **03**. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is NOT permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator.
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

: In words _____

Centre of Examination (in Capitals): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. When light propagates through a material medium of relative permittivity ϵ_r and relative permeability μ_r , the velocity of light, v is given by: (c - velocity of light in vacuum)

(1) $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$ (2) $v = c$
 (3) $v = \sqrt{\frac{\mu_r}{\epsilon_r}}$ (4) $v = \sqrt{\frac{\epsilon_r}{\mu_r}}$

2. An electric dipole of moment p is lying along a uniform electric field. The work done in rotating the dipole by 90° is:

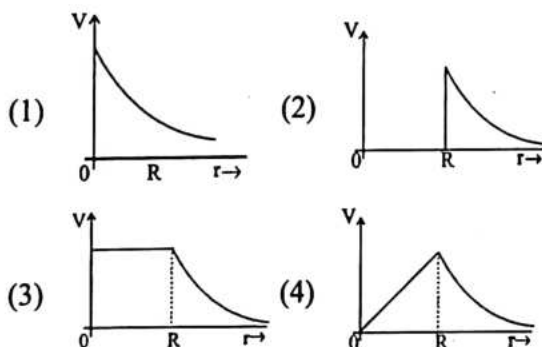
(1) $\sqrt{2}pE$ (2) $pE/2$
 (3) $2pE$ (4) pE

3. Electric potential at a point 'P' due to a point charge of $5 \times 10^{-9} \text{C}$ is 50V. The distance of 'P' from the point charge is

(Assume, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{N-m}^2\text{C}^{-2}$)

- (1) 90 cm
 (2) 3 cm
 (3) 0.9 cm
 (4) 9 cm

4. Which of the following correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the centre?



5. The unit of permittivity of free space ϵ_0 is:

- (1) Newton metre²/Coulomb²
 (2) Coulomb²/Newtonmetre²
 (3) Coulomb²/(Newton metre)²
 (4) Coulomb/Newton metre

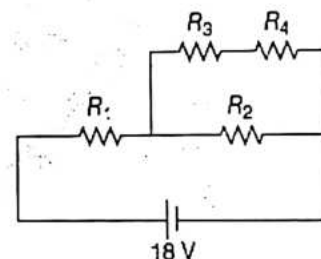
6. An aluminium wire is stretched to make its length, 0.4% larger. Then, percentage change in resistance is

- (1) 0.4% (2) 0.2%
 (3) 0.8% (4) 0.6%

7. Two identical heater filaments are connected first in parallel and then in series. At the same applied voltage, the ratio of heat produced in same time for parallel to series will be

- (1) 1 : 1 (2) 1 : 4
 (3) 2 : 1 (4) 4 : 1

8. In the given circuit, the internal resistance of the 18 V cell is negligible. If $R_1 = 400\Omega$, $R_3 = 100\Omega$ and $R_4 = 500\Omega$ and the reading of an ideal voltmeter across R_4 is 5V, then the value of R_2 will be



- (1) 550 Ω (2) 230 Ω
 (3) 300 Ω (4) 450 Ω

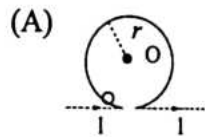
9. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The net resistance of their series combination is 1 k Ω . How much was the resistance on the left slot before interchanging the resistances?

- (1) 990 Ω (2) 505 Ω
 (3) 550 Ω (4) 910 Ω

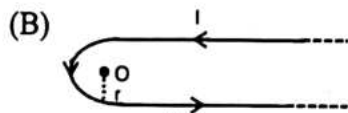
10. Match List-I with List-II.

List-I
(Current Configuration)

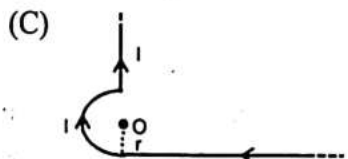
List-II
(Magnitude of Magnetic field at point O)



(I) $B_0 = \frac{\mu_0 I}{4\pi r} (\pi + 2)$



(II) $B_0 = \frac{\mu_0 I}{4 r}$



(III) $B_0 = \frac{\mu_0 I}{2\pi r} (\pi - 1)$



(IV) $B_0 = \frac{\mu_0 I}{4\pi r} (\pi + 1)$

Choose the correct answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
(2) A-III, B-IV, C-I, D-II
(3) A-I, B-III, C-IV, D-II
(4) A-II, B-III, C-IV, D-I

11. A long straight wire with a circular cross-section having radius R , is carrying a steady current I . The current I is uniformly distributed across this cross-section. Then, the variation of magnetic field due to current I with distance ($r < R$) from its centre will be

- (1) $B \propto r^2$ (2) $B \propto r$
(3) $B \propto \frac{1}{r^2}$ (4) $B \propto \frac{1}{r}$

12. An electron with energy 0.1 keV moves at right angle to the earth's magnetic field of $1 \times 10^{-4} \text{ Wbm}^{-2}$. The frequency of revolution of the electron will be (Take, mass of electron = $9.0 \times 10^{-31} \text{ kg}$)

- (1) $1.6 \times 10^5 \text{ Hz}$ (2) $5.6 \times 10^5 \text{ Hz}$
(3) $2.8 \times 10^6 \text{ Hz}$ (4) $1.8 \times 10^6 \text{ Hz}$

13. Two long straight wires P and Q carrying equal current 10 A each were kept parallel to each other at 4 cm distance. Magnitude of magnetic force experienced by 10 cm length of wire P is F_1 . If distance between wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be

- (1) $8F_1$ (2) $10F_1$
(3) $\frac{F_1}{10}$ (4) $\frac{F_1}{8}$

14. Given below are two statements.

Statement-I: If the number of turns in the coil of a moving coil galvanometer is doubled then the current sensitivity becomes double.

Statement-II: Increasing current sensitivity of a moving coil galvanometer by only increasing the number of turns in the coil will also increase its voltage sensitivity in the same ratio.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

15. The coercivity of a small magnet where the ferromagnet gets demagnetised is $3 \times 10^3 \text{ Am}^{-1}$. The current required to be passed in a solenoid of length 10 cm and number of turns 100, so that the magnet gets demagnetised when inside the solenoid is

- (1) 30 mA
(2) 60 mA
(3) 3 A
(4) 6 A

16. The total charge, induced in a conducting loop when it is moved in magnetic field depend on
- The rate of change of magnetic flux
 - Initial magnetic flux only
 - The total change in magnetic flux
 - Final magnetic flux only

17. The magnetic flux through a coil perpendicular to its plane is varying according to the relation, $\phi = (5t^3 + 4t^2 + 2t - 5)$ Wb. If the resistance of the coil is 5Ω , then the induced current through the coil at $t = 2$ s will be

- | | |
|------------|------------|
| (1) 15.6 A | (2) 16.6 A |
| (3) 17.6 A | (4) 18.6 A |

18. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: A bar magnet dropped through a metallic cylindrical pipe takes more time to come down compared to a non-magnetic bar with same geometry and mass.

Reason R: For the magnetic bar, Eddy currents are produced in the metallic pipe which oppose the motion of the magnetic bar.

In the light of the above statements, choose the correct answer from the options given below.

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is NOT the correct explanation of A.

19. The angle between the electric lines of force and the equipotential surface is:

- | | |
|-----------------|----------------|
| (1) 180° | (2) 0° |
| (3) 45° | (4) 90° |

20. A small square loop of side a and one turn is placed inside a larger square loop of side b and one turn ($b \gg a$). The two loops are coplanar with their centres coinciding. If a current I is passed in the square loop of side b , then the coefficient of mutual inductance between the two loops is

- | | |
|--|--|
| (1) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b}$ | (2) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{a}$ |
| (3) $\frac{\mu_0}{\pi} 8\sqrt{2} \frac{a^2}{b}$ | (4) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{b}$ |

21. In an LC oscillator, if values of inductance and capacitance become twice and eight times respectively, then the resonant frequency of oscillator becomes x times its initial resonant frequency.

The value of x is

- | | |
|------------|-----------|
| (1) $1/16$ | (2) 16 |
| (3) 4 | (4) $1/4$ |

22. In a series $L-C-R$ resonance circuit, if we change the resistance only, from a lower to higher value,

- the bandwidth of resonance circuit will increase
- the resonance frequency will increase
- the quality factor will increase
- the quality factor and the resonance frequency will remain constant

23. A circuit has a resistance of 12Ω and an impedance of 15Ω . The power factor of the circuit will be

- | | |
|----------|-----------|
| (1) 0.8 | (2) 0.4 |
| (3) 1.25 | (4) 0.125 |

24. The phase difference between the alternating current and emf is $\pi/2$. Which of the following cannot be the constituent of the circuit?

- | | |
|-------------|-------------|
| (1) C alone | (2) L, C |
| (3) R, L | (4) L alone |

25. For the plane electromagnetic wave given by $E = E_0 \sin(\omega t - kx)$ and $B = B_0 \sin(\omega t - kx)$, the ratio of average electric energy density to average magnetic energy density is

(1) 2 (2) $1/2$
(3) 1 (4) 4

26. The magnetic field of a plane electromagnetic wave is given by $B = 2 \times 10^{-8} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{j}$ T. The amplitude of the electric field would be

(1) 6 Vm^{-1} along X-axis
(2) 3 Vm^{-1} along Z-axis
(3) 6 Vm^{-1} along Z-axis
(4) $2 \times 10^{-8} \text{ Vm}^{-1}$ along Z-axis

27. In an electromagnetic wave, the electric field vector and magnetic field vector are given as $E = E_0 \hat{i}$ and $B = B_0 \hat{k}$, respectively. The direction of propagation of electromagnetic wave is along

(1) $\hat{k}$ (2) $\hat{j}$
(3) $(-\hat{k})$ (4) $(-\hat{j})$

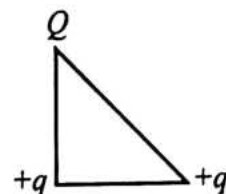
28. Electric field in a plane electromagnetic wave is given by $E = 50 \sin(500x - 10 \times 10^{10} t) \text{ V/m}$. The velocity of electromagnetic wave in this medium is (Given, c = speed of light in vacuum)

(1) $\frac{3}{2}c$ (2) c
(3) $\frac{2}{3}c$ (4) $\frac{c}{2}$

29. Three charges $+Q$, q and Q are placed respectively at distance 0, $d/2$ and d from the origin on the X-axis. If the net force experienced by $+Q$ placed at $x = 0$ is zero, then value of q is

(1) $\frac{+Q}{2}$ (2) $\frac{+Q}{4}$
(3) $\frac{-Q}{2}$ (4) $\frac{-Q}{4}$

30. Three charges Q , $+q$ and $+q$ are placed at the vertices of a right angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is



(1) $-2q$ (2) $\frac{-q}{1+\sqrt{2}}$
(3) $+q$ (4) $\frac{-\sqrt{2}q}{\sqrt{2}+1}$

31. A point dipole $\vec{p} = p_0 \hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the Y-axis at a distance d are, respectively [Take, $V = 0$ at infinity]

(1) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$
(2) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$
(3) $0, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$
(4) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$

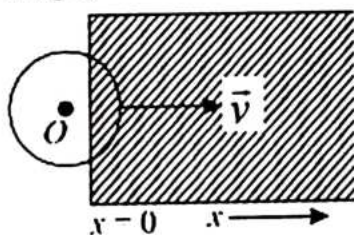
32. Equivalent resistance between the adjacent corners of a regular n -sided polygon of uniform wire of resistance R would be

(1) $\frac{(n-1)R}{n}$
(2) $\frac{(n-1)R}{(2n-1)}$
(3) $\frac{(n-1)R}{n^2}$
(4) $\frac{n^2 R}{n-1}$

33. If in a circular coil A of radius R , current I is flowing and in another coil B of radius $2R$ a current $2I$ is flowing, then the ratio of the magnetic fields, B_A and B_B produced by them will be
 (1) 1 (2) 2
 (3) $1/2$ (4) 4
34. A paramagnetic material has 10^{28} atoms/ m^3 . Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4} . Its susceptibility at 300 K is
 (1) 3.726×10^{-4} (2) 3.672×10^{-4}
 (3) 2.672×10^{-4} (4) 3.267×10^{-4}
35. An emf of 0.08 V is induced in a metal rod of length 10 cm held normal to a uniform magnetic field of 0.4 T , when moves with a velocity (perpendicular to length) of
 (1) 0.5 ms^{-1}
 (2) 2 ms^{-1}
 (3) 20 ms^{-1}
 (4) 3.2 ms^{-1}

Physics : Section-B (Q. No. 36 to 50)

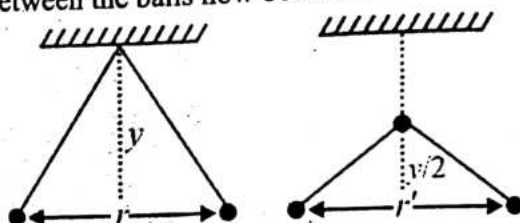
36. A constant magnetic field of 1 T is applied in the $x > 0$ region. A metallic circular ring of radius 1 m is moving with a constant velocity of 1 m/s along the X -axis. At $t = 0$, the centre of O of the ring is at $x = -1\text{ m}$. What will be the value of the induced emf in the ring at $t = 1\text{ s}$ (Assume the velocity of the ring does not change.)



- (1) 1 V (2) $2\pi\text{ V}$
 (3) 2 V (4) 0 V

37. An inductance coil has a resistance of $100\ \Omega$. When an AC signal of frequency 1000 Hz is applied to the coil, the applied voltage leads the current by 45° . The self-inductance of the coil is
 (1) $1.6 \times 10^{-2}\text{ H}$
 (2) $1.1 \times 10^{-1}\text{ H}$
 (3) $5.5 \times 10^{-5}\text{ H}$
 (4) $6.7 \times 10^{-7}\text{ H}$

38. Two pith balls carrying equal charges are suspended from a common point by strings of equal length, the equilibrium separation between them is r . Now the strings are rigidly clamped at half the height. The equilibrium separation between the balls now become:



- (1) $\left(\frac{2r}{3}\right)$
 (2) $\left(\frac{1}{\sqrt{2}}\right)^2$
 (3) $\left(\frac{r}{\sqrt{2}}\right)$
 (4) $\left(\frac{2r}{\sqrt{3}}\right)$

39. Two capacitors of capacities $2C$ and C are joined in parallel and charged up to potential V . The battery is removed and the capacitor of capacity C is filled completely with a medium of dielectric constant K . The potential difference across the capacitors will now be

- (1) $\frac{3V}{K+2}$ (2) $\frac{3V}{K}$
 (3) $\frac{V}{K+2}$ (4) $\frac{V}{K}$

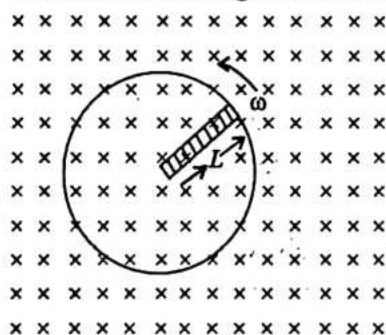
40. Two parallel infinite line charges with linear charge densities $+\lambda$ C/m and $-\lambda$ C/m are placed at a distance of $2R$ in free space. What is the electric field mid-way between the two line charges?

- (1) Zero
- (2) $\frac{2\lambda}{\pi\epsilon_0 R}$ N/C
- (3) $\frac{\lambda}{\pi\epsilon_0 R}$ N/C
- (4) $\frac{\lambda}{2\pi\epsilon_0 R}$ N/C

41. A charge of $4\mu\text{C}$ is to be divided into two parts. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be.

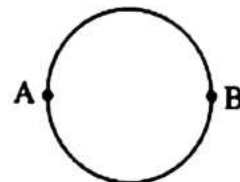
- (1) $1\mu\text{C}$ and $3\mu\text{C}$
- (2) $2\mu\text{C}$ and $2\mu\text{C}$
- (3) 0 and $4\mu\text{C}$
- (4) $1.5\mu\text{C}$ and $2.5\mu\text{C}$

42. A metallic rod of length L is rotated with an angular speed of ω normal to a uniform magnetic field B about an axis passing through one end of rod as shown in figure. The induced emf will be



- (1) $\frac{1}{2}BL^2\omega$
- (2) $\frac{1}{4}BL^2\omega$
- (3) $\frac{1}{2}B^2L^2\omega$
- (4) $\frac{1}{4}B^2L^2\omega$

43. A wire of resistance 12 ohms per meter is bent to form a complete circle of radius 10 cm. The resistance between its two diametrically opposite points, A and B as shown in the figure is:



- (1) 3Ω
- (2) $6\pi\Omega$
- (3) 6Ω
- (4) $0.6\pi\Omega$

44. A thin ring of radius R metre has charge q coulomb uniformly spread on it. The ring rotates about its axis with a constant frequency of f revolution/s. The value of magnetic induction in Wb/m^2 at the center of the ring is:

- (1) $\frac{\mu_0 q f}{2R}$
- (2) $\frac{\mu_0 q f}{2\pi R}$
- (3) $\frac{\mu_0 q}{2\pi f R}$
- (4) $\frac{\mu_0 q}{2 f R}$

45. According to Curie's law, the magnetic susceptibility of a paramagnetic material at an absolute temperature T is proportional to:

- (1) $1/T$
- (2) T
- (3) T^3
- (4) T^2

46. In an AC circuit an alternating voltage

$E = 200\sqrt{2} \sin 100t$ volts is connected to a capacitor of capacity $1 \mu\text{F}$. The r.m.s value of the current in the circuit is:

- (1) 10 mA
- (2) 100 mA
- (3) 200 mA
- (4) 20 mA

47. Two isolated metallic solid spheres of radii R and $2R$ are charged such that both have same charge density σ . The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ' . The ratio σ'/σ is
- (1) $5/3$ (2) $9/4$
(3) $4/3$ (4) $5/6$

48. Light with an energy flux of $25 \times 10^4 \text{ W/m}^2$ falls on a perfectly reflecting surface at normal incidence. If the surface area is 15cm^2 , the average force exerted on the surface is:
- (1) $1.25 \times 10^{-6} \text{ N}$
(2) $2.50 \times 10^{-6} \text{ N}$
(3) $1.20 \times 10^{-6} \text{ N}$
(4) $3.0 \times 10^{-6} \text{ N}$

49. A coil in the shape of an equilateral triangle of side L is suspended between the pole pieces of a permanent magnet such that B is in plane of the coil. If due to a current i in the triangle a torque τ acts on it, the side L of the triangle is:

- (1) $\frac{2}{\sqrt{3}} \left(\frac{\tau}{Bi} \right)$ (2) $2 \left[\frac{\tau}{\sqrt{3}Bi} \right]^{1/2}$
(3) $\frac{2}{\sqrt{3}} \left[\frac{\tau}{Bi} \right]^{1/2}$ (4) $\frac{1 \times \tau}{\sqrt{3}Bi}$

50. A galvanometer of resistance, G , is shunted by a resistance S ohm. To keep the main current in the circuit unchanged, the resistance to be put in series with the galvanometer is:

- (1) $\frac{S^2}{(S+G)}$
(2) $\frac{SG}{(S+G)}$
(3) $\frac{G^2}{(S+G)}$
(4) $\frac{G}{(S+G)}$

Chemistry : Section-A (Q. No. 51 to 85)

51. Homogeneous mixture of two or more than two components is called;

- (1) Solute
(2) Solvent
(3) Solution
(4) None of these

52. In a galvanic cell electron flow will be from;

- (1) negative electrode to positive electrode.
(2) positive electrode to negative electrode.
(3) there will be no flow of electrons.
(4) cathode to anode in the external circuit.

53. For the reaction, $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, rate is expressed as;

- (1) $\frac{1}{3} \frac{d[\text{N}_2]}{dt} = -\frac{1}{2} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$
(2) $-\frac{d[\text{N}_2]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$
(3) $-\frac{d[\text{N}_2]}{dt} = -\frac{3d[\text{H}_2]}{dt} = \frac{2d[\text{NH}_3]}{dt}$
(4) $-\frac{d[\text{N}_2]}{dt} = -\frac{d[\text{H}_2]}{dt} = \frac{d[\text{NH}_3]}{dt}$

54. Anomalous behavior of nitrogen is due to;

- (1) Small size and high electronegativity
(2) Non availability of d-orbital in valence shell
(3) Ease of multiple bond formation
(4) All of these

55. Among the following series of transition metal ions, the one where all metal ions have $3d^5$ configuration is;

- (1) $\text{Cr}^{2+}, \text{Mn}^{2+}, \text{Fe}^{2+}, \text{Co}^{3+}$
(2) $\text{Cr}^{+}, \text{Mn}^{2+}, \text{Fe}^{3+}, \text{Co}^{4+}$
(3) $\text{Cr}^{2+}, \text{Mn}^{3+}, \text{Fe}^{4+}, \text{Co}^{5+}$
(4) $\text{Cr}^{3+}, \text{Mn}^{3+}, \text{Fe}^{3+}, \text{Co}^{3+}$

56. Which of the following can be termed as mixed complex?

- (1) $K_4[Fe(CN)_6]$
- (2) $[Cu(NH_3)_4]SO_4$
- (3) $[Co(NH_3)_4(NO_2)Cl]Cl$
- (4) K_2FeO_4

57. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Ice melts slower at high altitudes.

Reason (R): At high altitudes atmospheric pressure is high.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

58. Which of the following electrolytes is not preferred in a salt bridge?

- (1) KCl
- (2) KNO_3
- (3) NH_4NO_3
- (4) NaCl

59. Match List-I with List-II

List-I (Reaction type)	List-II (Unit of rate constant (k))
A. First order reaction	I. $\text{mol L}^{-1} \text{s}^{-1}$
B. Second order reaction	II. $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
C. Third order reaction	III. $\text{mol}^{-1} \text{L s}^{-1}$
D. Zero order reaction	IV. s^{-1}

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-I, C-III, D-IV
- (4) A-II, B-III, C-IV, D-I

60. Which of the following trihalides of nitrogen is least basic?

- (1) NF_3
- (2) NCl_3
- (3) NBr_3
- (4) NI_3

61. Match List-I with List-II

List-I (Lanthanoids)	List-II (Outer electronic configuration)
A. Lanthanum	I. $4f^4 5d^1 6s^2$
B. Gadolinium	II. $5d^1 6s^2$
C. Cerium	III. $4f^7 5d^1 6s^2$
D. Lutetium	IV. $4f^1 5d^1 6s^2$

Choose the correct answer from the options given below:

- (1) A-IV; B-III; C-II; D-I
- (2) A-III; B-II; C-I; D-IV
- (3) A-II; B-III; C-IV; D-I
- (4) A-I; B-II; C-III; D-IV

62. A complex involving dsp^2 hybridisation has;

- (1) Square planar geometry
- (2) A tetrahedral geometry
- (3) An octahedral geometry
- (4) Trigonal planar geometry

63. Calculate the volume by volume percentage of a solution of 15 mL of alcohol in 60 mL of water:

- (1) 20%
- (2) 25%
- (3) 30%
- (4) 50%

64. The Nernst equation giving dependence of electrode oxidation potential on concentration is:

- (1) $E = E^\circ + \frac{2.303RT}{nF} \log[M^{n+}]$
- (2) $E = E^\circ - \frac{2.303RT}{nF} \log \frac{[M^{n+}]}{[M]}$
- (3) $E = E^\circ - \frac{2.303RT}{nF} \log[M^{n+}]$
- (4) $E = E^\circ + \frac{2.303RT}{nF} \log \frac{[M]}{[M^{n+}]}$

65. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Hydrolysis of ethyl acetate is a pseudo first order reaction.

Reason (R): Water is present in large excess and therefore its concentration remain constant during the reaction.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

66. Inert pair effect is shown by;

- (1) s-block elements
- (2) p-block elements
- (3) d-block elements
- (4) f-block elements

67. Find the **incorrect** statement:

- (1) The IE of d-block element is greater than s-block and lesser than p-block elements
- (2) Greater the IE, more stable is the formed O.S.
- (3) First IE of d-block elements varies in the group as $3d > 4d < 5d$
- (4) None of these

68. Which of the following is **not** an ambidentate ligand?

- (1) NO_2^-
- (2) SCN^-
- (3) CN^-
- (4) NH_3

69. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Osmosis does not take place in two isotonic solutions separated by a semipermeable membrane.

Reason (R): Isotonic solutions have the same osmotic pressure.

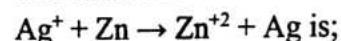
In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

70. The standard oxidation potential E° for the half reactions are:



The standard EMF of the cell



- (1) 1.56 V
- (2) -1.56 V
- (3) -0.01 V
- (4) 0.01 V

71. For the following statements:

- A. The simple chemical reactions which occur only in one step are called elementary reactions.
- B. The molecularity of a reaction is a whole number.
- C. In a complex reaction, the slowest elementary step is the rate determining step.

The correct statements is/are;

- (1) A only
- (2) A and B
- (3) A, B and C
- (4) A and C

72. Which of the following compounds is paramagnetic?
 (1) N_2O_4 (2) NO_2
 (3) Cl_2O_7 (4) N_2O
73. The number of unpaired electrons in Fe^{3+} ($Z = 26$) are;
 (1) 4 (2) 5
 (3) 6 (4) 3
74. In SCN ligand if N is attached to central atom, the name of ligand is;
 (1) Thiocyanato-N (2) Cyanato-N
 (3) Thiocyanato-S (4) Cyanato-S
75. Which of the following concentration factors is affected by change in temperature?
 (1) Molarity (2) Molality
 (3) Mole fraction (4) mass percentage
76. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Cu^{2+} ions get reduced more easily than H^+ ions.
Reason (R): Standard electrode potential of Cu^{2+}/Cu is 0.34V.
 In the light of the above statements, choose the correct answer from the options given below:
 (1) A is false but R is true.
 (2) A is true but R is false.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
77. Which of the following rate laws has an overall order of 0.5 for the reaction involving substances x, y and z?
 (1) $\text{Rate} = k[\text{C}_x][\text{C}_y][\text{C}_z]$
 (2) $\text{Rate} = k[\text{C}_x]^{0.5}[\text{C}_y]^{0.5}[\text{C}_z]^{0.5}$
 (3) $\text{Rate} = k[\text{C}_x]^{1.5}[\text{C}_y]^{-1}[\text{C}_z]^0$
 (4) $\text{Rate} = k[\text{C}_x][\text{C}_y]^{-2}[\text{C}_z]^0$
78. Given below are two statements:
Statement-I: Conc. H_2SO_4 can be used for drying H_2S gas.
Statement-II: Conc. H_2SO_4 shows very good dehydrating property.
 In the light of the above statements, choose the correct answer from the options given below:
 (1) Statement-I is true but Statement-II is false.
 (2) Statement-I is false but Statement-II is true.
 (3) Both Statement-I and Statement-II are true.
 (4) Both Statement-I and Statement-II are false.
79. Which of the following transition element shows the highest oxidation state?
 (1) Mn (2) Fe
 (3) V (4) Sc
80. Primary and secondary valency of Pt in $[\text{Pt}(\text{en})_2\text{Cl}_2]$ are;
 (1) +4 and 4
 (2) +4 and 6
 (3) +6 and 4
 (4) +2 and 6
81. Four gases like H_2 , He, CH_4 and CO_2 has Henry's constant values (K_H) are 69.16, 144.979, 0.413 and 1.67. The gas which is more soluble in liquid is;
 (1) He (2) CH_4
 (3) H_2 (4) CO_2
82. The standard reduction potential values of three metallic cations X, Y and Z are 0.52, -3.03 and -1.18 V respectively. The order of reducing power of the corresponding metal is;
 (1) $\text{Y} > \text{Z} > \text{X}$
 (2) $\text{X} > \text{Y} > \text{Z}$
 (3) $\text{Z} > \text{Y} > \text{X}$
 (4) $\text{Z} > \text{X} > \text{Y}$

83. The units of rate constant and rate of a reaction are identical for;
- (1) zero order reaction.
 - (2) first order reaction.
 - (3) second order reaction.
 - (4) third order reaction.
84. M.P (melting point) of the following hydrides follow in the order;
- (1) $\text{NH}_3 > \text{SbH}_3 > \text{AsH}_3 > \text{PH}_3$
 - (2) $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 > \text{NH}_3$
 - (3) $\text{SbH}_3 > \text{AsH}_3 < \text{PH}_3 < \text{NH}_3$
 - (4) $\text{SbH}_3 < \text{AsH}_3 < \text{PH}_3 < \text{NH}_3$
85. Ti^{2+} is purple while Ti^{4+} is colourless, because
- (1) There is no crystal field effect in Ti^{4+} .
 - (2) Ti^{2+} has $3d^0$ configuration.
 - (3) Ti^{4+} has $3d^0$ configuration.
 - (4) Ti^{4+} is a very small cation when compared to Ti^{2+} and hence does not absorb any radiation.

Chemistry : Section-B (Q. No. 86 to 100)

86. The formula of the complex hexaamminecobalt (III) chloride sulphate is;
- (1) $[\text{Co}(\text{NH}_3)_6]\text{ClSO}_4$
 - (2) $[\text{Co}(\text{NH}_3)_6\text{Cl}]\text{SO}_4$
 - (3) $[\text{Co}(\text{NH}_3)_6]\text{ClSO}_4$
 - (4) None of these
87. An ideal solution is that which?
- (1) Obey Raoult's law
 - (2) Shows positive deviation from Raoult's law
 - (3) Shows negative deviation from Raoult's law
 - (4) Has no connection with Raoult's law
88. Specific conductance of 0.1 M nitric acid is $6.3 \times 10^{-2} \text{ ohm}^{-1} \text{ cm}^{-1}$. The molar conductance of solution;
- (1) $630 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$
 - (2) $315 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$
 - (3) $100 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$
 - (4) $6300 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$

89. Given below are two statements:
Statement-I: Increase in concentration of reactant will not change the rate for a zero-order reaction.
Statement-II: Rate constant for a zero-order reaction is a constant for a particular initial concentration.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) Statement-I is correct but Statement-II is incorrect.
 - (2) Statement-I is incorrect but Statement-II is correct.
 - (3) Both Statement-I and Statement-II are correct.
 - (4) Both Statement-I and Statement-II are incorrect.
90. Which of the following has the highest bond energy?
- (1) O – O
 - (2) S – S
 - (3) Se – Se
 - (4) Te – Te
91. Which of the following oxide is basic?
- (1) CrO
 - (2) Cr_2O_3
 - (3) CrO_3
 - (4) Cr_2O_4
92. Choose the species from the following carrying a single lone pair on central atom Xenon.
- XeF_5^+
 - XeO_3
 - XeO_2F_2
 - XeF_5^-
 - XeO_3F_2
 - XeOF_4
 - XeF_4
- (1) (B), (C) and (E)
 - (2) (E), (F) and (G)
 - (3) (A), (B), (C) and (F)
 - (4) (A), (B), (D), (E) and (F)
93. The order of osmotic pressure of isomolar solution of BaCl_2 , NaCl and sucrose is;
- (1) $\text{BaCl}_2 > \text{NaCl} > \text{sucrose}$
 - (2) $\text{NaCl} > \text{BaCl}_2 > \text{sucrose}$
 - (3) $\text{sucrose} > \text{NaCl} > \text{BaCl}_2$
 - (4) $\text{BaCl}_2 > \text{sucrose} > \text{NaCl}$

94. In $H_2 - O_2$ fuel cell the reaction occurring at cathode is;

- (1) $2H_2O + O_2 + 4e^- \rightarrow 4OH^-$
- (2) $2H_2 + O_2 \rightarrow 2H_2O(l)$
- (3) $H^+ + OH^- \rightarrow H_2O$
- (4) $H^+ + 1e^- \rightarrow \frac{1}{2} H_2$

95. For a zero order reaction;

- (1) $t_{\frac{1}{2}} \propto a$
- (2) $t_{\frac{1}{2}} \propto \frac{1}{a}$
- (3) $t_{\frac{1}{2}} \propto a^2$
- (4) $t_{\frac{1}{2}} \propto \frac{1}{a^2}$

96. Which of the following statement regarding double salt and complex salt is not correct?

- a. Double salts dissociate into simple ions completely when dissolved in water.
- b. Carnallite is an example of complex salt.
- c. Complex salt do not dissociate into simple ions when dissolved in water.
- d. Potassium ferrocyanide is an example of double salt.

- (1) (a) and (b)
- (2) (b), (c) and (d)
- (3) (b) and (d)
- (4) (a) and (c)

97. Lanthanoids and actinoids differ but not in;

- (1) tendency to show more compounds in +3 O.S.
- (2) tendency to form oxocations.
- (3) basicity of hydroxides.
- (4) Atomic or ionic size contraction.

98. Select correct pair of co-ordination compounds in which first complex compound/ion is more stable than second compound/ion?

- (1) $[Ni(CN)_4]^{2-}$, $[Ni(NH_3)_6]^{2+}$
- (2) $[PdBr_2(NH_3)_2]$, $[PdBr(ONO)(NH_3)_2]$
- (3) $[Co(SCN)_4]^{2-}$, $[CoF_6]^{3-}$
- (4) $[Cr(NH_3)_2(en)_2]^{3+}$, $[Cr(en)_3]^{3+}$

99. Van't Hoff factor of $Ca(NO_3)_2$ is;

- (1) 1
- (2) 2
- (3) 3
- (4) 4

100. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): If the activation energy of a reaction is zero, temperature will have no effect on the rate constant.

Reason (R): Lower the activation energy, faster is the reaction.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

Botany : Section-A (Q. No. 101 to 135)

101. The repressor of the *lac* operon is synthesised from the;

- (1) *z* gene.
- (2) *y* gene.
- (3) *a* gene.
- (4) *i* gene.

102. Choose the correct pair of contrasting traits of pod colour in pea plant studied by Mendel in his experiments.

- (1) Violet-dominant; White-recessive.
- (2) Yellow-dominant; Green-recessive
- (3) Green-dominant; Yellow-recessive
- (4) White-dominant; Violet-recessive

103. ZW-ZZ type of sex determination is observed in;

- (1) humans.
- (2) birds.
- (3) *Drosophila*.
- (4) grasshoppers.

104. Identify the **correct** statement w.r.t the monohybrid cross experiment conducted by Mendel in pea plant.

- (1) Mendel cross-pollinated the tall F_1 plants to obtain F_2 generation.
- (2) In the F_1 generation some of the offspring were dwarf.
- (3) The character of the parents that was not seen in the F_1 generation was expressed in F_2 generation.
- (4) The proportion of plants that were dwarf was $3/4^{\text{th}}$ of the F_2 plants.

105. In 4 o'clock plant, the gene for red flower colour (R) is incompletely dominant over the gene for white flower colour (r), hence the plants heterozygous for flower colour have pink flowers (Rr). What will be ratio of offsprings in a cross between two pink flowered plant?

- (1) 75% red flowers, 25% pink flowers
- (2) All red flowers
- (3) 50% red flowers, 50% pink flowers
- (4) Red : Pink : White :: 1 : 2 : 1

106. In angiosperms, identify the cells that have ploidy n, 2n and 3n, respectively.

- (1) Antipodal; Egg cell; Endosperm
- (2) Egg cell; Nucellus; Endosperm
- (3) Endosperm; Nucellus; Egg cell
- (4) Antipodals; Synergids; Integuments

107. In a majority of aquatic plants such as water hyacinth and water lily, pollination takes place by;

- (1) water currents.
- (2) biotic agents only.
- (3) abiotic agents only.
- (4) insects or wind.

108. A couple has four children with blood types O, A, AB, and B, respectively. The possible genotypes of couple could be;

- (1) $I^A i$ and $I^B i$
- (2) $I^A i$ and ii
- (3) $I^B I^B$ and $I^A I^A$
- (4) I^{AA} and I^{Bi}

109. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

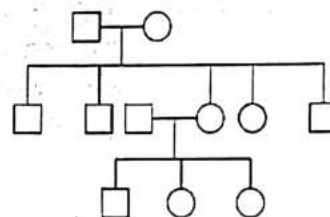
Assertion (A): A recessive character that was not expressed in heterozygous condition may be expressed again when it becomes homozygous.

Reason (R): The recessive characters are only expressed in homozygous condition.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

110. The figure given below is the representative pedigree analysis of a/an;



- (1) autosomal recessive trait.
- (2) sex-linked recessive disorder.
- (3) sex-linked dominant disorder.
- (4) autosomal dominant trait.

111. Match the columns and find out the correct combination:

Column-I	Column-II
(a) ACU	(i) Asn
(b) AAC	(ii) Thr
(c) AGG	(iii) Arg
(d) GAG	(iv) Glu
(1) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)	
(2) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)	
(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)	
(4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)	

112. Given below are two statements:

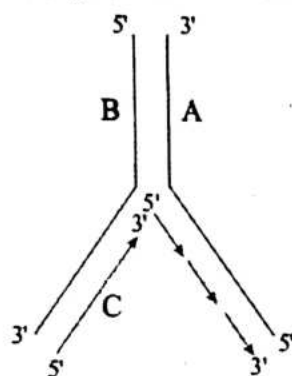
Statement-I: Sequence annotation is the approach of HGP that focuses on identifying all the genes that are expressed in RNA.

Statement-II: Expressed Sequence Tags (ESTs) is the approach of HGP that focuses on simply sequencing the whole set of genome that contained all the coding and non-coding sequence.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

113. A replication fork is shown in the figure. Identify the template DNA.



- (1) Both A and B
- (2) Both B and C
- (3) C only
- (4) Both B and C

114. The outermost and innermost wall layers of microsporangium in an anther, respectively, are;

- (1) endothecium and tapetum.
- (2) epidermis and endodermis.
- (3) epidermis and tapetum.
- (4) epidermis and middle layer.

115. Identify the **incorrectly** matched pair.

- (1) Incomplete dominance - Inheritance of flower colour in snapdragon flower
- (2) Co-dominance - Inheritance of flower colour in dog flower
- (3) Complete dominance- Height of pea plant
- (4) Multiple allelism - ABO Blood group system

116. The codon which has dual functions is ____.

- (1) UAA
- (2) AUG
- (3) UGA
- (4) UAG

117. _____ used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome.

- (1) Alfred Sturtevant
- (2) Sutton and Boveri
- (3) Thomas Hunt Morgan
- (4) de Vries, Correns and von Tschermak

118. The process of translation of the mRNA to protein begins when the 'A' subunit of the ribosome encounters a 'B'. A and B, respectively, are;

- (1) larger, mRNA.
- (2) smaller, tRNA.
- (3) larger, tRNA.
- (4) smaller, mRNA.

119. In grass family, the cotyledon is known as;

- (1) scutellum.
- (2) coleorhiza.
- (3) epiblast.
- (4) endocarp.

120. All of the following are transcribed by RNA polymerase I in eukaryotes, except;

- (1) 5.8S rRNA.
- (2) 28S rRNA.
- (3) 18S rRNA.
- (4) 5srRNA.

121. Out of the given diseases, how many is/are not Mendelian disorder(s)?

Haemophilia, Down's syndrome, Cystic fibrosis, Phenylketonuria, Turner syndrome

- (1) One (2) Two
(3) Three (4) Four

122. Which of the following crosses correctly justifies the test cross?

- (a) $Ww \times ww$ (b) $WW \times ww$
(c) $Ww \times Ww$ (d) $WW \times WW$

Choose the correct option(s):

- (1) (a) and (c) (2) (b) and (d)
(3) (a) and (b) (4) (c) and (d)

123. All the following options are true for VNTRs, except that they/their;

- (1) show very low degree of polymorphism.
(2) are a mini-satellite DNA.
(3) used as a probe in DNA fingerprinting.
(4) size ranges from 0.1 to 20 kb.

124. Match List-I with List-II.

List-I (Part of gynoecium)	List-II (Post-fertilisation structure)
(A) PEN	(I) Embryo
(B) Zygote	(II) Endosperm
(C) Ovule	(III) Fruit
(D) Ovary	(IV) Seed

Choose the correct answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-I, B-II, C-IV, D-III
(4) A-II, B-I, C-III, D-IV

125. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): Histones are organised to form a unit of eight molecules called histone octamer.

Reason (R): Histones are rich in the basic amino acid residues lysine and arginine.

In the light of the above statements, choose the correct answer from the options given below:

- (1) (A) is true but (R) is false.
(2) (A) is false but (R) is true.
(3) Both (A) and R are true and (R) is the correct explanation of A.
(4) Both (A) and (R) are true but (R) is NOT the correct explanation of A.

126. All of the following are correct w.r.t agents of pollination, except;

- (1) pollination by wind is more common amongst biotic pollinations.
(2) pollination by water is quite rare in flowering plants and is limited to about 30 genera, mostly monocotyledons.
(3) majority of flowering plants use a range of animals as pollinating agents.
(4) majority of insect-pollinated flowers are large, colourful, fragrant and rich in nectar.

127. Match List-I with List-II.

List-I	List-II
(A) Colour blindness	(I) Is inborn error of metabolism
(B) Sickle-cell anaemia	(II) Quantitative problem of synthesising too few globin molecules
(C) Phenylketonuria	(III) Occurs in about 8 percent of males and only about 0.4 percent of females
(D) Thalassemia	(IV) Transmitted when both the parents are carrier for the gene

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
(2) A-II, B-IV, C-III, D-I
(3) A-I, B-II, C-IV, D-III
(4) A-IV, B-I, C-II, D-III

128. Given below are two statements:

Statement-I: Pollen grains are well preserved as fossils because of the presence of sporopollenin.

Statement-II: The intine of pollen grain is made up of sporopollenin.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

129. Identify the **odd** one w.r.t ploidy of given cells.

- (1) Nucellus
- (2) Integuments
- (3) Synergids
- (4) Megaspore mother cell

130. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): Cleistogamous flowers produce assured seed-set even in the absence of pollinators.

Reason (R): In cleistogamous flowers, the anthers and stigma lie close to each other. When anthers dehisce in the flower buds, pollen grains come in contact with the stigma to effect pollination.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

131. A typical angiosperm anther has ___ lobes and a total of _____ theca.

- (1) two; two
- (2) two; four
- (3) four; four
- (4) one; two

132. If both male and female flowers are present on the same plant such as castor and maize (monoecious), it prevents;

- (1) both autogamy and geitonogamy.
- (2) both geitonogamy and xenogamy.
- (3) geitonogamy but not xenogamy.
- (4) autogamy but not geitonogamy.

133. For conducting artificial hybridization, if the mother plant produces unisexual flowers;

- (1) There is a need of emasculation and bagging.
- (2) There is no need of emasculation and bagging.
- (3) There is a need of emasculation but no need of bagging.
- (4) There is no need of emasculation but bagging is needed for female plants.

134. Pick the **correct** statement(s) w.r.t filiform apparatus and choose the **correct** option.

- A. Present at the chalazal end of embryo sac.
 - B. Present as special thickenings at the micropylar tip in synergids.
 - C. Guides the pollen tube into synergids.
- (1) A and B
 - (2) B and C
 - (3) only B
 - (4) only C

135. Match List-I with List-II.

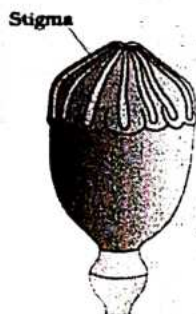
List-I	List-II
(A) Tassels of a corn cob	(I) Produce two types of flowers
(B) <i>Oxalis</i>	(II) Flower is about 6 feet in height
(C) <i>Yucca</i>	(III) Trap pollen grains
(D) <i>Amorphophallus</i>	(IV) Gets pollinated by moth

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-I, C-IV, D-II

Botany : Section-B (Q. No. 136 to 150)

136. Choose the correct option representing the figure given below.

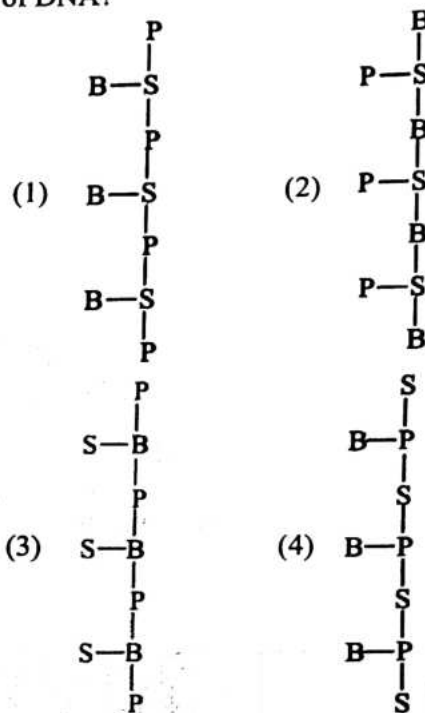


- (1) Multicarpellary, apocarpous gynoecium of *Michelia*.
- (2) Multicarpellary, syncarpous pistil of *Papaver*.
- (3) Multicarpellary, syncarpous gynoecium of *Michelia*.
- (4) Multicarpellary, apocarpous pistil of *Papaver*.

137. At which part of the soluble RNA does the amino acid bind?

- (1) 5' end
- (2) anticodon loop
- (3) 3' end
- (4) Both 5' and 3' ends

138. Which of the following shows the correct positions of the phosphate (P), sugar (S) and Nitrogen base (B) molecules in the line structure of DNA?



139. Which of the following is true for the double helical model of DNA?

- (1) The two chains of DNA have anti-parallel polarity.
- (2) Adenine binds to thymine by 3 hydrogen bonds.
- (3) The backbone is constituted by sugar-phosphate, and the bases project outside.
- (4) The pitch of the helix is 34 nm.

140. The mRNA sequence generated from the coding strand with the sequence given below will be;

5' ATGTCGTAGCTTGCATGAC 3'

- (1) 5' AUGUCGUAGCUUGCAUGAC 3'
- (2) 3' ATGTCGTAGCTTGCATGAC 5'
- (3) 3' AUGUCGUAGCUUGCAUGAC 5'
- (4) 5' ATGTCGTAGCTTGCATGAC 3'

141. Identify the **incorrect** statement.

- (1) Depending on base composition (A:T rich or G:C rich), length of segment, and number of repetitive units, satellite DNA is classified in micro-satellites and mini-satellites etc.
- (2) The technique of DNA fingerprinting was initially developed by Alec Jeffreys.
- (3) Polymorphisms are non-inheritable from parents to children.
- (4) DNA fingerprinting is the basis of paternity testing, in case of disputes.

142. Which law of inheritance is used to explain the expression of only one of the parental characters in a monohybrid cross in the F_1 and the expression of both in the F_2 generation?

- (1) Law of dominance
- (2) Law of segregation
- (3) Law of independent assortment
- (4) None of these

143. Total population of 800 individuals which formed F_2 population (9:3:3:1) of a cross between yellow round and green wrinkled. Find the number of plants with yellow and wrinkled seeds

- | | |
|---------|---------|
| (1) 150 | (2) 400 |
| (3) 800 | (4) 300 |

144. Given below are two statements:

Statement-I: In 1902, de Vries, Correns and von Tschermak independently rediscovered Mendel's results on the inheritance of characters.

Statement-II: By 1900, the chromosome movement during meiosis had been worked out. In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

145. Following are the stages in the formation of pollen grains during the process of microsporogenesis:

- (a) Cells of sporogenous tissue undergo meiotic divisions
- (b) Formation of microspore tetrads
- (c) Development into pollen grains
- (d) Dissociation of microspores from the tetrads
- (e) Dehiscence of anther to release pollen grains

Choose the correct sequence of these stages from the options given below:

- (1) (a)-(b)-(d)-(c)-(e)
- (2) (b)-(a)-(c)-(d)-(e)
- (3) (c)-(a)-(b)-(d)-(e)
- (4) (d)-(b)-(a)-(c)-(e)

146. Pedigree analysis is **not** possible for:

- (1) Cystic fibrosis.
- (2) Sickle-cell anaemia.
- (3) Thalassaemia.
- (4) Turner's syndrome.

147. Which of the following are components of a mature non-albuminous seed?

- (a) Perisperm
- (b) Endosperm
- (c) Embryo axis
- (d) Cotyledons
- (e) Seed coat

Choose the correct answer from the options given below:

- (1) (c), (d), and (e) only
- (2) (b), (c), and (d) only
- (3) (a), (b), (c), and (d) only
- (4) (a), (c), (d), and (e) only

148. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): The inborn error of metabolism in humans is inherited as the autosomal recessive trait.

Reason (R): In affected individuals, phenylalanine is accumulated and converted into phenylpyruvic acid and other derivatives.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

149. A hybrid DNA molecule having ^{15}N (heavy isotope of nitrogen) in one strand and ^{14}N in the other was allowed to replicate in a medium containing ^{14}N only. After 4 generations, how many DNA molecules will have hybrid DNA?

- (1) One
- (2) Two
- (3) Fifteen
- (4) Zero

150. Match List-I with List-II.

List-I	List-II
(A) Tailing	(I) Terminate the process of transcription
(B) Capping	(II) Initiate the process of transcription
(C) Sigma factor	(III) Adding methyl guanosine triphosphate
(D) Rho factor	(IV) Adding adenylate residues

Choose the correct answer from the options given below:

- (1) A-IV, B-II, C-I, D-III
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-III, C-II, D-I
- (4) A-III, B-I, C-IV, D-II

Zoology : Section-A (Q. No. 151 to 185)

151. What are the characteristics of an ideal contraceptive?

- (1) User-friendly, easily available, effective, and reversible
- (2) Difficult to use, less available, effective, and reversible
- (3) User-friendly, easily available, ineffective, and irreversible
- (4) Difficult to use, less available, ineffective, and irreversible

152. Match List-I with List-II:

List-I	List-II
A. Mons pubis	I. Partially covers the opening of vagina
B. Labia majora	II. Fleshy folds of tissue
C. Hymen	III. Lies at upper junction of labia minora
D. Clitoris	IV. Cushion of fatty tissue

Choose the correct answer from the options given below:

- (1) A-IV, B-II, C-I, D-III
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-IV, C-I, D-III

153. The theory of spontaneous generation believed that:

- (1) life arose from living forms only.
- (2) life can arise from both living and non-living.
- (3) life can arise from decaying and rotting matter.
- (4) life comes only from pre-existing life.

154. Which is the correct sequence of male accessory ducts starting from the testis?

- (1) Rete testis $\rightarrow$ vasa efferentia $\rightarrow$ epididymis $\rightarrow$ vas deferens
- (2) Rete testis $\rightarrow$ vas deferens $\rightarrow$ vasa efferentia $\rightarrow$ epididymis
- (3) Rete testis $\rightarrow$ vas deferens $\rightarrow$ epididymis $\rightarrow$ vasa efferentia
- (4) Rete testis $\rightarrow$ vas deferens $\rightarrow$ vasa efferentia $\rightarrow$ epididymis

155. Match List-I with List-II:

List-I	List-II
A. 500 mya	I. Sea weeds existed
B. 350 mya	II. Invertebrates were formed and active.
C. 320 mya	III. Some of the land reptiles went back into water to evolve into fish-like reptiles.
D. 200 mya	IV. Jawless fishes evolved

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-IV, C-II, D-III
- (4) A-III, B-I, C-II, D-IV

156. Read the following statements.

- A. The middle piece possesses numerous mitochondria.
- B. The human male ejaculates about 200 to 300 million sperms during a coitus.
- C. A plasma membrane envelops the whole body of sperm.
- D. Mitochondria produces energy for the movement of tail that facilitates sperm motility essential for gametogenesis.

Choose the option with **correct** statements.

- (1) A only
- (2) B and D only
- (3) A and D only
- (4) A, B and C only

157. Evidence of evolution from fossils is known as:

- (1) paleontological evidence.
- (2) embryological evidence.
- (3) morphological evidence.
- (4) All of these

158. Read the following statements.

- A. Oogenesis is initiated during the embryonic development stage.
- B. In spermatogenesis, one primary spermatocyte can form only one spermatid.
- C. After implantation, finger-like projections appear on the inner cell mass called chorionic villi.
- D. The milk produced during the initial few days of lactation is called colostrum.

Choose the option with **correct** statements.

- (1) A, C and D only
- (2) B, C and D only
- (3) C and D only
- (4) A, B and D only

159. Given below are two statements:

Statement-I: Scrotum is the only male reproductive part which is included in male external genitalia.

Statement-II: The trophoblast layer attaches to endometrium.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

160. Match List-I with List-II:

List-I	List-II
A. Mammary lobes	I. 60,000-80,000 in each ovary
B. Primary follicles left at puberty	II. 15-20 in each breast
C. Secondary spermatocytes	III. 250 in each testis
D. Testicular lobules	IV. 23 chromosomes

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-I, C-IV, D-III

161. The original variety of Darwin's finches had a beak specialized in:

- (1) seed-eating.
- (2) insect-eating.
- (3) fruit-eating.
- (4) cactus-eating.

162. Given below are two statements:

Statement-I: The fitness, according to Darwin, refers ultimately and only to reproductive fitness.

Statement-II: Individuals who better fit in an environment, leave less progeny than others.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

163. Match List-I with List-II:

List-I	List-II
A. Reproductive tract infection	I. Induced abortion
B. Medical termination of pregnancy	II. Progestasert
C. Copper releasing IUD	III. Chlamydisis
D. Hormone releasing IUD	IV. Multiload 375

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-IV, C-I, D-III

164. Which of the following are included in female external genitalia?

- | | |
|-----------------|-----------|
| A. Mons pubis | B. Cervix |
| C. Labia majora | D. Uterus |
- (1) A, B and C only (2) A and C only
(3) C and D only (4) D only

165. Arrange the periods of the Mesozoic era from early to recent on a geological time scale.

- (1) Triassic → Jurassic → Cretaceous
- (2) Jurassic → Triassic → Cretaceous
- (3) Triassic → Cretaceous → Jurassic
- (4) Cretaceous → Triassic → Jurassic

166. Match List-I with List-II:

List-I	List-II
A. By the end of second trimester	I. Embryo's heart is formed.
B. By the end of nine months	II. Body is covered with fine hair.
C. After one month of pregnancy	III. Foetus develops limbs and digits.
D. By the end of the second month	IV. Foetus is fully developed.

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-IV, C-I, D-III
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-IV, C-II, D-I

167. Given below are two statements: one is labeled as Assertion (A) and the other is labeled as Reason (R).
Assertion (A): Not all copulations lead to pregnancy.

Reason (R): Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

168. Which natural contraceptive method involves the male partner withdrawing his penis from the vagina just before ejaculation?

- (1) Periodic abstinence
- (2) Coitus interruptus
- (3) Lactational amenorrhea
- (4) Surgical methods

169. In humans, the male accessory glands secrete seminal plasma which is rich in:

- (1) fructose, calcium and no enzymes.
- (2) fructose, calcium and certain enzymes.
- (3) sucrose, calcium and no enzymes.
- (4) sucrose, calcium and certain enzymes.

170. Match List-I with List-II:

List-I	List-II
A. Medical termination of pregnancy legalized in	I. Barrier method
B. Family planning initiated in	II. 1951
C. Vault	III. Natural method
D. Lactational amenorrhea	IV. 1971

Choose the correct answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-IV, B-II, C-I, D-III
- (4) A-IV, B-II, C-III, D-I

171. Spermiogenesis is the transformation of:

- (1) spermatogonium into primary spermatocyte.
- (2) spermatogonium into functional spermatozoa.
- (3) primary spermatocytes into secondary spermatocytes.
- (4) spermatids into spermatozoa.

172. Given below are two statements: one is labeled as Assertion (A) and the other is labeled as Reason (R).

Assertion (A): Evolution is a directed process in the sense of determinism.

Reason (R): Evolution is based on chance events in nature and chance mutations in the organisms. In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

173. Which one of the following events is correctly matched with the time period in a normal menstrual cycle?

- (1) Release of secondary oocyte: 5th day
- (2) Endometrium regenerates: 5th – 10th day
- (3) Endometrium ready for implantation: 11th day
- (4) Rise in progesterone level: 1st – 15th day

174. Match List-I with List-II:

List-I	List-II
A. Pre-historic cave art	I. 10,000 years back
B. Agriculture	II. 18,000 years back
C. Neanderthal man	III. 1.5 million years ago
D. Fossils discovered in Java	IV. 1,00,000–40,000 years back

Choose the correct answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-II, C-III, D-I

175. The layer of uterus which exhibits strong contractions during delivery of the baby is/are:

- (1) perimetrium.
- (2) myometrium.
- (3) endometrium.
- (4) both myometrium and endometrium.

176. Which technique involves directly injecting a sperm into an ovum in the laboratory?

- (1) Zygote intrafallopian transfer
- (2) Intrauterine transfer
- (3) Gamete intrafallopian transfer
- (4) Intra-cytoplasmic sperm injection

177. Match List-I with List-II:

List-I

List-II

- | | |
|-----------------|-----------------------------|
| A. Ampulla | I. Finger like projections |
| B. Infundibulum | II. Widest part of the tube |
| C. Fimbriae | III. Part closer to ovary |
| D. Isthmus | IV. Last part of oviduct |

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-III, C-I, D-IV
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-IV, C-I, D-III

178. Given below are two statements:

Statement-I: Lichens can be used as industrial pollution indicators.

Statement-II: The geological history of Earth closely correlates with the biological history of Earth.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

179. What is the purpose of amniocentesis?

- (1) For diagnosis of genetic disorders before birth.
- (2) To increase the fertility of women.
- (3) For reducing menstrual problems.
- (4) To provide contraception.

180. Which of the following defines Hardy-Weinberg principle?

- (1) $p^2 + 2pq + q^2 = 0$
- (2) $q^2 + p^2 + 2pq = 3$
- (3) $p^2 + 2pq + q^2 = 1$
- (4) $p^2 + 3pq + q^2 = 1$

181. Match List-I with List-II:

List-I

List-II

- | | |
|------------------------|--------------------------------|
| A. Giant pteridophytes | I. Moth |
| B. Primitive earth | II. Spores |
| C. Panspermia | III. Reducing atmosphere |
| D. Industrial melanism | IV. Fell to form coal deposits |

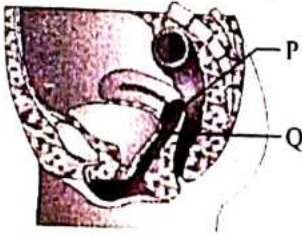
Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-III, C-II, D-I

182. Which of the following is related to the Spotted Cuscus?

- (1) *Lemur*
- (2) Numbat
- (3) Marsupial mole
- (4) Anteater

183. Identify the parts P and Q respectively.



- (1) P - vagina; Q - cervix
- (2) P - cervix; Q - vagina
- (3) P - uterus; Q - cervix
- (4) P - cervix; Q - external genitalia

184. Match List-I with List-II:

List-I	List-II
A. Provide nutrition to spermatogonia	I. Male germ cells
B. Sperm formation	II. Vasa efferentia
C. Synthesise androgens	III. Sertoli cells
D. Male sex accessory ducts	IV. Leydig cells

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-III, C-I, D-IV
- (3) A-III, B-I, C-IV, D-II
- (4) A-IV, B-II, C-III, D-I

185. Which of the following is the **correct** sequence of events in evolution?

- I. Formation of non-cellular aggregates.
 - II. Synthesis of organic molecules (amino acids, sugars, nitrogen bases, pigments and fats).
 - III. Formation of multi-cellular life forms.
 - IV. Formation of single-celled organisms.
- (1) I → III → II → IV
 - (2) II → I → IV → III
 - (3) II → III → IV → I
 - (4) I → II → IV → III

Zoology : Section-B (Q. No. 186 to 200)

186. Choose the **incorrect** statement.

- (1) The cells of alveoli secrete milk, which is stored in the cavities (lumens) of alveoli.
- (2) Lactiferous duct opens into mammary ampulla.
- (3) The alveoli open into mammary tubules.
- (4) A functional mammary gland is characteristic of all female mammals.

187. Match List-I with List-II:

List-I	List-II
A. Secondary oocyte	I. 8-16 celled embryo formed in fallopian tube
B. Zygote	II. Implantation occurs in the uterine endometrium
C. Morula	III. Released by the ovary
D. Blastocyst	IV. Diploid cell formed as a result of fertilisation

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-III, B-IV, C-I, D-II

188. Which of the following are **incorrect**?

- A. Amphibians evolved into reptiles.
- B. Homology in the vertebrate brain indicates common ancestry.
- C. Mutations are directional.
- D. The rate of appearance of new forms is not linked to evolution.

Choose the **most appropriate** answer from the options given below:

- (1) A and C only
- (2) B and C only
- (3) B and D only
- (4) C and D only

189. Match List-I with List-II:

List-I	List-II
A. Uterus	I. Spermatogenesis
B. Vagina	II. Oogenesis
C. Testes	III. Implantation
D. Ovaries	IV. Insemination

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-IV, C-II, D-III
- (4) A-III, B-II, C-IV, D-I

190. Given below are two statements:

Statement-I: Hymen is a reliable indicator of sexual experience.

Statement-II: In women, the hymen is always torn during the first coitus.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

191. Match List-I with List-II:

List-I	List-II
A. Coelacanth	I. The thorn of <i>Bougainvillea</i> and tendrils of <i>Cucurbita</i>
B. Dinosaurs	II. Disappeared about 65 mya
C. Analogous organs	III. Caught in South Africa in 1938
D. Homologous organs	IV. Flippers of Penguins and Dolphins

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-IV, C-I, D-III

192. Select the correct statement(s).

- (A) First menstrual cycle begins at puberty is called menopause.
 - (B) Each seminiferous tubule is lined on its inside by male germ cells only.
 - (C) For normal fertility, at least 60 percent of sperms must have normal shape and size.
 - (D) The menstrual flow results due to breakdown of endometrial lining of the uterus.
- (1) A and B only
 - (2) C and D only
 - (3) C only
 - (4) D only

193. Match List-I with List-II:

List-I	List-II
A. Saltation	I. Total genes and their alleles in a population
B. Gene pool	II. Changes in allele frequencies in the population due to chance alone.
C. Founder effect	III. Single step large mutation
D. Genetic drift	IV. When a small group of population separates from the main population and forms a new colony.

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-I, D-IV
- (3) A-I, B-IV, C-III, D-II
- (4) A-II, B-IV, C-III, D-I

194. Given below are two statements:

Statement-I: The use of condom can help prevent the transmission of syphilis and gonorrhea.

Statement-II: Hepatitis-B cannot be transmitted from an infected mother to the foetus.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

195. Match List-I with List-II:

List-I	List-II
A. Darwin	I. Abiogenesis
B. Oparin	II. Use and disuse of organs
C. Lamarck	III. Mutation
D. Hugo de Vries	IV. Evolution by natural selection

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-IV, C-II, D-III

196. Following are the statements with reference to Saheli.

- A. It is an oral contraceptive for males.
- B. It is a 'once a week' pill.
- C. It was developed at the Central Drug Research Institute (CDRI) in Lucknow.
- D. It has a limited contraceptive value but several side effects.

Choose the **correct** statements from the options given below:

- (1) B and C only
- (2) C and D only
- (3) B and D only
- (4) A and C only

197. Match List-I with List-II:

List-I	List-II
A. Stabilizing selection	I. 650-800 cc
B. Directional selection	II. 900 cc
C. <i>Homo erectus</i>	III. More individuals acquire value other than mean character value
D. <i>Homo habilis</i>	IV. More individuals acquire mean character value

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

198. Match List-I with List-II:

List-I	List-II
A. Menstrual phase	I. Graafian follicle ruptures to release the ovum
B. Follicular phase	II. Corpus luteum secretes progesterone
C. Ovulation	III. Follicular development and endometrium repair
D. Luteal phase	IV. Endometrium breakdown and released from uterus

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-III, C-I, D-II

199. Read the following statements:

- A. The skull of an adult chimpanzee is more like an adult human skull than a baby chimpanzee skull.
- B. The first mammals were like shrews.
- C. Due to continental drift, North American animals were overridden by South American fauna.
- D. About 200 mya, the first cellular forms of life appeared on earth.

How many of the above statement(s) is/are incorrect?

- (1) Two
- (2) Three
- (3) Four
- (4) One

200. Match List-I with List-II:

List-I	List-II
A. Pills	I. Intrauterine device
B. Condoms	II. Prevent ovulation
C. Vasectomy	III. Prevent sperm from reaching the cervix
D. CuT	IV. Semen contains no sperm

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-IV, B-III, C-II, D-I